

COLLOID

COMPARISON OF LYOPHILIC AND LYOPHOBIC SOLS

Lyophilic Sols	Lyophobic Sols
1. Prepared by direct mixing with dispersion medium. 2. Little or no charge on particles. 3. Particles generally solvated. 4. Viscosity higher than dispersion medium; set to a gel. 5. Precipitated by high concentration of electrolytes. 6. Reversible. 7. Do not exhibit Tyndall effect. 8. Particles migrate to anode or cathode, or not at all.	1. Not prepared by direct mixing with the medium. 2. Particles carry positive or negative charge. 3. No solvation of particles. 4. Viscosity almost the same as of medium; do not set to a gel. 5. Precipitated by low concentration of electrolytes. 6. Irreversible. 7. Exhibit Tyndall effect. 8. Particles migrate to either anode or cathode.

colloid

A **colloid** is a heterogeneous mixture where one substance (dispersed phase) is evenly distributed in another substance (dispersion medium) at a microscopic level. The particle size of the dispersed phase typically ranges between 1 and 1000 nanometers. Colloidal particles are too small to settle under gravity but large enough to scatter light (Tyndall effect).

Types of Colloids

Colloids are classified based on the physical state of the dispersed phase and the dispersion medium. Below are the main types:

1. **Sol:** A colloidal system with a solid dispersed in a liquid or gas. Example: Paints, muddy water.
2. **Gel:** A colloid with a liquid dispersed in a solid. Example: Jelly, gelatin.
3. **Aerosol:** A colloid where the dispersed phase is a solid or liquid, and the dispersion medium is a gas. Example: Fog (liquid in gas), smoke (solid in gas).
4. **Emulsion:** A colloid of two immiscible liquids, where one liquid is dispersed in the other. Example: Milk, mayonnaise.

5. **Foam:** A colloid with gas dispersed in a liquid or solid.Example: Shaving cream (gas in liquid), foam rubber (gas in solid).
6. **Solid Sol:** A colloid where solid particles are dispersed in a solid medium.Example: Colored glass, alloys.
7. **Solid Foam:** A colloid where gas is dispersed in a solid.Example: Pumice, Styrofoam.

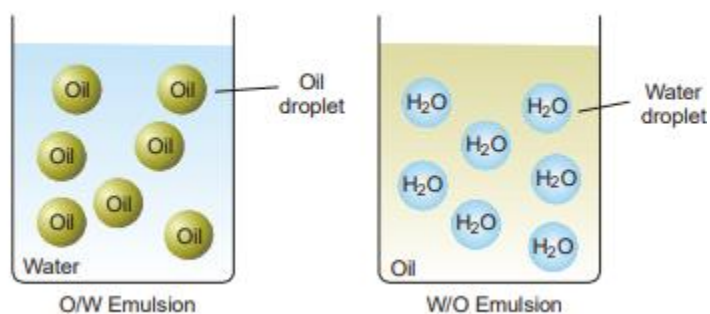
Examples of Colloidal Systems

Dispersed Phase	Dispersion Medium	Type of Colloid	Example
Solid	Liquid	Sol	Paint, ink
Liquid	Liquid	Emulsion	Milk, butter
Gas	Liquid	Foam	Soap lather, whipped cream
Solid	Gas	Aerosol	Smoke, dust in air
Liquid	Gas	Aerosol	Fog, clouds
Gas	Solid	Solid Foam	Foam rubber, sponge
Solid	Solid	Solid Sol	Alloys, gemstones

EMULSIONS

These are liquid-liquid colloidal systems. In other words, an emulsion may be defined as a dispersion of finely divided liquid droplets in another liquid. Generally one of the two liquids is water and the other, which is immiscible with water, is designated as oil. Either liquid can constitute the dispersed phase. There are two types of emulsions.

(a) Oil-in-Water type (O/W type) ; (b) Water-in-Oil type (W/O type)



Examples of Emulsions

- (1) Milk is an emulsion of O/W type. Tiny droplets of liquid fat are dispersed in water.
- (2) Stiff greases are emulsions of W/O type, water being dispersed in lubricating oil.

Preparation of Emulsions

The dispersal of a liquid in the form of an emulsion is called emulsification. This can be done by agitating a small proportion of one liquid with the bulk of the other. It is better accomplished by passing a mixture of the two liquid through a colloid mill known as homogenizer. The emulsions obtained simply by shaking the two liquids are unstable. The droplets of the dispersed phase coalesce and form a separate layer. To have a stable emulsion, small amount of a third substance called the Emulsifier or Emulsifying agent is added during the preparation. This is usually a soap, synthetic detergent, or a hydrophilic colloid.

COMPARISON OF PHYSICAL ADSORPTION AND CHEMISORPTION

Physical adsorption	Chemisorption
1. Caused by intermolecular van der Waal's forces. 2. Depends on nature of gas. Easily liquefiable gases are adsorbed readily. 3. Heat of adsorption is small (about 5 kcal mol⁻¹). 4. Reversible. 5. Occurs rapidly at low temperature; decreases with increasing temperature. 6. Increase of pressure increases adsorption; decrease of pressure causes desorption. 7. Forms multimolecular layers on adsorbent surface.	1. Caused by chemical bond formation. 2. Much more specific than physical adsorption. 3. Heat of adsorption is large (20–100 kcal mol⁻¹). 4. Irreversible. 5. Increases with increase of temperature. 6. Change of pressure has no such effects. 7. Forms unimolecular layer.

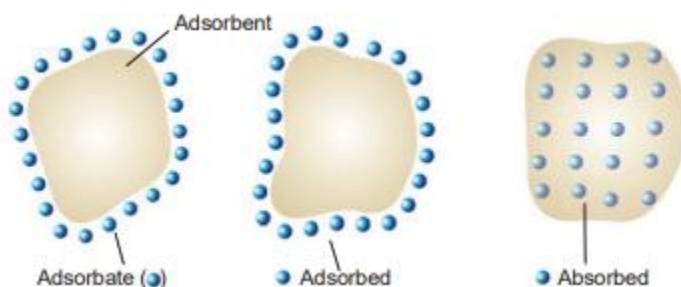
SOL

Sols are colloidal systems in which a solid is dispersed in a liquid. These can be subdivided into two classes :

- (a) Lyophilic sols (solvent-loving) (b) Lyophobic sols (solvent-hating)

Adsorption

The phenomenon of concentration of molecules of a gas or liquid at a solid surface is called adsorption. The substance that deposits at the surface is called Adsorbate and the solid on whose surface the deposition occurs is called the Adsorbent



Examples of Adsorption

(1) **Adsorption of a dye by a charcoal.** If finely divided charcoal is stirred into a dilute solution of methylene blue (an organic dye), the depth of colour of the solution decreases appreciably. The dye molecules have been adsorbed by charcoal particles.

(2) **Adsorption of a gas by charcoal.** If a gas (SO_2 , Cl_2 , NH_3) is treated with powdered charcoal in a closed vessel, the gas pressure is found to decrease. The gas molecules concentrate on charcoal surface and are said to be adsorbed.

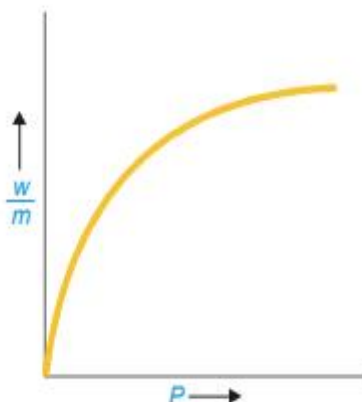
Isotherm with plots.

Freundlich Adsorption Isotherm

Freundlich proposed an empirical relation in the form of a mathematical equation.

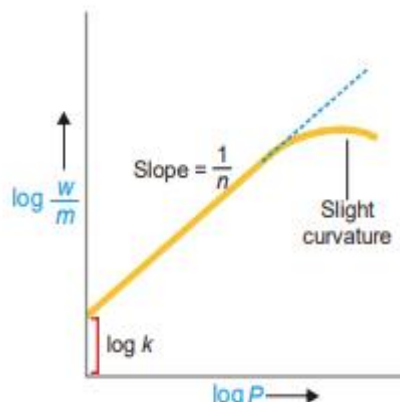
$$\frac{w}{m} = k P^{1/n}$$

where w is the mass of the gas adsorbed on a mass m of adsorbent at a pressure P ; k and n are constants depending on the nature of the gas and the adsorbent and on temperature. This relation is generally represented in the form of a curve obtained by plotting the mass of the gas adsorbed per unit mass of adsorbent (w/m) against equilibrium pressure.



■ **Figure 23.6**

Freundlich isotherm; a plot of mass of adsorbed gas per unit mass of adsorbent.



■ **Figure 23.7**

Plot of $\log w/m$ against $\log P$ shows slight curvature at higher pressures.

Freundlich isotherm is not applicable at high pressures. Taking logarithms on both sides of Freundlich equation, we have

$$\log \frac{w}{m} = \log k + \frac{1}{n} \log P$$

This is equation for a straight line. Thus a plot of $\log (w/m)$ against $\log P$ should be a straight line with slope $1/n$ and intercept $\log k$. However, it is actually found that the plots were straight lines at low pressures, while at higher pressure they showed a slight curvature, especially at low temperature. This indicated that Freundlich equation is approximate and does not apply to adsorption of gases by solids at higher pressures.

LANGMUIR ADSORPTION ISOTHERM

Langmuir (1916) derived a simple adsorption isotherm based on theoretical considerations. It was named after him.

Assumptions

Langmuir made the following assumptions.

- (1) The layer of the gas adsorbed on the solid adsorbent is one-molecule thick.
- (2) The adsorbed layer is uniform all over the adsorbent.
- (3) There is no interaction between the adjacent adsorbed molecules.